Selected S-E-P: clearance and a concrete recovery-source choice

15 October 2026 — bounded analytical review, not a traffic experiment

Status. The user selects S-E-P's priority direction, closing S-E-A2. The previously proposed cooperative and state/work formulas remain trial laws. Two explicit recovery-source laws are formed below for comparison; neither is selected. Stop at **SEP-R1**, Type 2 G3. Global hybrid well-posedness, complete MP6 and Step 3 are not established.

1. Priority selection and the E-clearance boundary

Retain $P_j = F_j(t - b/v)$, $q_j = P_j - Y_j$, $N_j = F_j - Y_j$, eligible arrivals $\lambda_j = \dot{P}_j$, and initially empty terminal histories. The independent B lane has $Y_B = P_B$. For the common triangular family,

$$c(u) = \frac{wk_j u}{w+u}, \quad C(v) = c(v), \quad U(f) = \frac{wf}{wk_j - f}, \quad 0 < R \le C(v_-), \quad v \in [v_-, v_+]. \tag{1}$$

Write $B(t)$ for shared available work, distinct from the B lane index. S-E-P gives unqueued A first claim to λ_A when it fits B. Whenever A is queued, use the cooperative branch, including an empty E offer. Thus cooperation does not disappear solely because E clears.

An exact local clearance diagnostic illustrates the distinction. Take

$$B = 4/5, \quad \lambda_A = 3/5, \quad \lambda_E = 1/10, \quad q_A > 0, \quad q_E \downarrow 0, \quad c_A = c_E = 1, \quad C \ge B. \tag{2}$$

The unit-cost limit is an accounting diagnostic, not a primary parameter choice or an H1 reachability claim. Immediately before E clears, both queues can receive $(g_A, g_E) = (2/5, 2/5)$, so E drains. At E clearance, the selected cooperative template serves

$$(g_A, g_E) = (7/10, 1/10), \qquad (\dot{q}_A, \dot{q}_E) = (-1/10, 0). \tag{3}$$

A literal "strict A priority unless both queues are positive" toggle instead assigns $g_E = 0$ at $q_E = 0$, and $g_E = 2/5$ at $q_E > 0$. It would require

$$\dot{q}_E = 1/10 \ (q_E = 0), \qquad \dot{q}_E = -3/10 \ (q_E > 0). \tag{4}$$

There is no nonnegative absolutely continuous continuation from zero: on any positive component (a, b), integration gives $q_E(t) = -3(t-a)/10 < 0$; staying identically zero contradicts (4). An averaged/sliding rule would be another law, not the literal toggle. Equation (3) is already the proposed S-E-P behavior. No new latch, smoothing, positive queue floor or time-step convention is adopted.

2. A speed-marked finite recovery profile

Keep the prior local work family for conditional qualification,

$$c_A = 1, \qquad c_E = 1 + \delta + \beta \frac{g_A}{C(v)} \frac{[u_A - u_E]_+}{v}, \quad \delta, \beta \geq 0, \quad g_A + c_E g_E \leq B. \tag{5}$$

The prior note proves fixed-regime unique queued roots under the sufficient condition $1 + \delta - \beta(1 + v_+/w) > 0$. Free trace marks use $u_j = v$; queued trace marks use $u_j = U(g_j)$. Those representative speed and current-work identifications remain hypotheses, not measured kinematics.

Let $a_r > 0$ be a common physical recovery acceleration and $m_{\text{ref}} = 1$ vehicle-equivalent fix normalization, not a packet size. For an insertion-speed mark $u \geq 0$, define

$$\begin{aligned} u_R &= U(R), & V(\theta; u) &= \sqrt{(w+u)^2 + 2wa_r\theta} - w, \\ D(\theta; u) &= [R - c(V(\theta; u))]_+, & \theta &\geq 0. \end{aligned} \tag{6}$$

Here θ is boundary-observed age, including backward propagation, as in the earlier isolated recovery derivation. The positive part is the proposed physical loss law, not a numerical clipping repair. The profile is zero if $u \geq u_R$, and otherwise ends after

$$\begin{aligned} \Theta(u) &= \frac{u_R - u}{a_r} + \frac{u_R^2 - u^2}{2a_r w}, \qquad \Theta_{\max} = \Theta(0), \\ W(u) &= \int_0^\infty D(\theta; u) d\theta = \frac{k_c(u_R)[u_R - u]_+^2}{2a_r}, \qquad k_c(u) = \frac{wk_j}{w+u}. \end{aligned} \tag{7}$$

Thus $0 \leq D \leq R$, it decreases with age and speed mark, and $W(u) \leq W_0 := W(0) < \infty$. On the active part, differentiating (6) gives $|\partial_u D| \leq k_j$ and $|\partial_\theta D| \leq a_r k_j$. These bounds extend across the continuous zero-loss cutoff in the Lipschitz sense.

The isolated profile is a deduction using the point moving-bottleneck case in Leclercq et al. (2016). Summing profiles from continuous mass is an additional aggregate overlap assumption. The source's native wave/void interactions are not reproduced exactly. In particular, it does not show that every slow A departure should create a fresh independent profile.

3. Two concrete laws for the same shared account

Both candidates retain the same S-E-P allocation, trial work/trace laws, empty initial history, common parameters and one budget output:

$$B(t) = [R - p(t)]_+. \tag{8}$$

For each admitted stream, define a proposed speed-marked contribution

$$p_j(t) = \frac{1}{m_{\text{ref}}} \int_0^t D(t-s; u_j(s))\, dY_j(s) = \frac{1}{m_{\text{ref}}} \int_0^t D(t-s; u_j(s)) g_j(s) ds, \quad j = A, E. \tag{9}$$

R-E: E-origin disturbance, A follows the existing restriction. Set $p = p_E$. Actual admitted E quantity and its predicted speed determine new impairment. A affects that speed and admission through the coupled node and consumes the resulting shared service. Its slow discharge is not separately a fresh source. This is not the old fixed-speed M1 kernel: its mark varies with current predicted traffic state and actual admission.

R-AE: both slow streams renew impairment. Set $p = p_E + p_A$. A passage with mark below u_R also deposits a new profile, even when its low speed resulted from a prior E restriction. That is an explicit renewal/independence assumption. The kernel is neutral for sufficiently fast passage, not triggered merely by a queue flag.

These are specified constitutive candidates once parameters and the trial node are supplied, not names for unspecified future research. Their physical speed marks, overlap and complete event proofs are unfinished. R-AE's stronger assumption may count genuine successive acceleration burdens, or may overcount followers already constrained by one disturbance. R-E deliberately omits such independent follower renewal. Both limitations require evidence; neither follows from conservation alone.

Internal source histories combine into one p and one B. There is no separate service ceiling per past passage, integer admission, hidden initial speed oracle or longitudinal recovery zone. Only the last $\Theta_{\max}$ of predicted admission/mark history can contribute. This is recovery memory in the point account, not S-V approach-speed ODEs.

For a *fixed* admissible mode schedule, the prior monotone inverse makes g_j and the trace marks locally Lipschitz in B. Equations (6),(9) then give a causal Volterra map with difference bounded by $K\Delta t\|\Delta B\|_\infty$ on a short interval with identical past. Together with the 1-Lipschitz map in (8), this is a contraction for small enough proof interval. No numerical time scale is introduced. It gives local fixed-mode causal construction, not a theorem through arbitrary simultaneous or grazing queue events or a full proof of the hybrid model.

4. Exact recovery-tail contrast and finite-work bounds

Suppose no E is admitted after T_E. Under R-E,

$$p_E(t) = 0 \quad (t \geq T_E + \Theta_{\max}), \qquad \dot{p}_E \leq 0 \quad (t > T_E) \tag{10}$$

almost everywhere. A can still be waiting during that recovery. Queue clearance never resets p artificially.

Consider an A-only queued tail, no new eligible arrivals, and $0 < B < R$. S-E-P gives $g_A = B$, and its trial queued speed mark is $U(B)$. Because $c(U(B)) = B$,

$$D(0; U(B)) = R - B. \tag{11}$$

R-AE therefore adds new A loss at rate

$$\frac{g_A D(0; u_A)}{m_{\text{ref}}} = \frac{B(R-B)}{m_{\text{ref}}} > 0; \qquad R = 1,\ B = 3/4,\ m_{\text{ref}} = 1 \implies 3/16. \tag{12}$$

R-E has no such new source. For identical inherited history at a comparison instant, (12) is the difference in initial loss derivatives. Old profiles also recover, so R-AE's total loss need not increase; it may recover more slowly. No global ordering of two endogenously different trajectories is claimed. The rational tail case is an exact local diagnostic, not a new primary initialization or an H1 reachability certificate.

For either candidate, nonnegative kernels and mass conservation imply by Tonelli

$$\int_0^\infty p(t)dt \leq \begin{cases} M_E W_0/m_{\text{ref}}, & \text{R-E}, \\ (M_A + M_E)W_0/m_{\text{ref}}, & \text{R-AE}. \end{cases} \tag{13}$$

M_j denotes total mass eventually admitted from its terminal stream; early-merged traffic belongs to A and is not forced to merge again. The bound also holds before full clearance using the upper bound M. Since $R - B = \min(R, p) \leq p$, (13) establishes a finite integrated service loss for these particular laws, conditional on an admissible solution.

With $c_{\max} = 1 + \delta + \beta$, all mass eligible by $t_* = L/v$, and $M_{\mathcal{S}} = M_E$ or $M_A + M_E$, the prior work-conserving bound now yields

$$T_{\text{clear}} \leq t_* + \frac{c_{\max}(q_A(t_*) + q_E(t_*)) + M_{\mathcal{S}} W_0/m_{\text{ref}}}{R}. \tag{14}$$

Thus finite loss is no longer an unspecified assumption for either trial source law. This does not prove existence, hybrid uniqueness or empirical fidelity. No positive service floor or inherited M1 uniqueness proof is used.

5. A limiting case that constrains the traffic claim

Take an initially neutral account, empty A queue, constant A eligible rate $\lambda_A = a < R$, and the selected free-A priority branch. While A remains free, both recovery laws have no A source because $u_A = v \geq u_R$. Using $c_0 = 1 + \delta$, the instantaneous work law gives

$$0 \leq g_E \leq \frac{B - a}{c_0}. \tag{15}$$

For the nonincreasing recovery profiles, differentiating (9) gives

$$\dot{p}_E = \frac{g_E(t) D(0; u_E(t))}{m_{\text{ref}}} + \frac{1}{m_{\text{ref}}} \int_0^t \partial_\theta D(t - s; u_E(s)) g_E(s) ds \leq \frac{R}{m_{\text{ref}}} g_E. \tag{16}$$

On this branch $B > 0$ and $B = R - p_E$. Set $x = B - a$. Then

$$\dot{x} \geq -\frac{R}{m_{\text{ref}} c_0} x, \qquad x(t) \geq (R - a) \exp\left[-\frac{Rt}{m_{\text{ref}} c_0}\right] > 0. \tag{17}$$

Hence A cannot begin queueing in finite time in this constant-A branch. E may wait and reduce shared service, but its own admission is throttled before the smooth loss crosses A's protected offer. Adding the R-AE source does not create a first A queue there: its extra source is still neutral until A actually becomes slow under the trial rule.

This is a specific limiting result, not an external-seed requirement or a theorem that primary A delay is impossible. H1 produces time-varying $\lambda_A = v(r - e(t - b/v))$ over its arrival pulse, generally increasing; for varying $a(t)$ the inequality contains an additional $-\dot{a}$ term. The constant-rate barrier therefore does not apply directly. A consequential joined A-impact/MP6 witness remains to be established rather than assumed.

Unchanged interfaces. For the retained all-action domain $M < k_j b$ and $\dot{F}_j \leq C(v)$, terminal vacancy stays positive and H1 is protected for any eventual admissible departures. Physical mass is $N = F - Y$, and complete local TST remains

$$J(v) = \frac{M(L - \ell/2)}{v} + \int_0^\infty (q_A + q_E + q_B) dt. \tag{18}$$

Memory after completion is not vehicle-time. Scalar reconstruction and gradient usefulness do not follow from these bounds; neither an interior optimum nor full-lane-state necessity is required.

Gate SEP-R1. Do ordinary slow A departures merely follow the represented disturbance (R-E), or renew it through new loss profiles (R-AE)? Both are concrete source-law candidates; both need physical and global-event qualification. Recommend R-E first as the minimal primary candidate, not as proof independent A recovery is negligible. Neither source law is selected. S-E-P stays selected and the whole review/Step 3 remains open.